

Currents Program

Bringing water currents out of the isolated Lab and into the actual game — as a tested, opt-in system that never touches the shipped build.

Project **Floatsam** Branch **currents-program** Date **2026-07-11** PR **#2 open**

C1-C6 BUILT · DEFAULT-OFF

§1 Executive summary

We answered one product question — *how do currents enter the game?* — and shipped a full, tested implementation skeleton across six phases on an isolated branch.

- **The design question is resolved, grounded in the code.** The scrolling game fixes the player's x, so a whole-screen field only ever touches one vertical column. The everyday unit is therefore a **localized cell** that scrolls in like a gate; a whole-screen field is a **rare Surge** event.
- **Zero risk to the shipped game.** Currents are opt-in and default-off; a bit-identical test proves the off path is byte-for-byte the old behavior. Classic's golden master and the 80,000-run fairness matrix are untouched.
- **Fairness is guaranteed by construction.** Cells and surge share one budget clamp anchored to the weakest creature. The weakest reference bot still clears 8 gates through currents ~98% of the time.
- **Every design fork is a tunable variable**, per direction — nothing hard-committed.

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PHASES BUILT (C1-C6)

156

TESTS GREEN (+44)

+2,214

LINES, 19 NEW FILES

0

CHANGES TO SHIPPED GAME

Recommendation: merge `currents-program` → `main` (zero-risk, unlocks the Catalog & Sandbox at the live URL), then prioritize a player-facing entry point and the full fairness sweep before promoting currents from dev-reachable to public.

§2 The core decision

The engine has exactly one extension point: `FluidField.sampleForce(x, y, t)` returns an acceleration summed into the body. In the scrolling game the player's *x* is fixed, so a screen-anchored field is sampled at one column forever — a beautiful 2-D current collapses to a 1-D wobble.

So currents that you *travel through* must live in world coordinates and move. That reframes the whole feature: **localized current cells are the grammar; a rare whole-screen Surge is the punctuation.** The owner's instinct to keep the whole-screen effect rare is now the architecture, not a preference.

§3 What was built

PHASE	DELIVERABLE	TESTS
C1	Five pure shape primitives: Capsule (finite lane), Radial (boil/drain), Wedge (ramp), Ring (annulus), GradientBand (shear)	18
C2	20-pattern catalog (helpful/hostile + scope + game-clearance) & a browsable free-swim gallery	7
C3	Scrolling <code>CurrentSpawner</code> + <code>ScrollingCurrentField</code> adapter (pooled, seeded, world→local, single clamp)	9
C4	Cells opt-in in Power Dive + in-play telegraph + fairness spot check	3
C5	Rare whole-screen Surge (envelope scheduler + field, shares the cell clamp)	7
C6	Currents Sandbox — every design fork a live slider	—

§4 Architecture

The seam

Currents plug into `sampleForce` and nowhere else — `FluidBody`, the game loop, and all locked Classic files are untouched.

Cells

A cell is a localized field built once in a local frame ($x = 0$), placed at a `worldX` that scrolls left. `ScrollingCurrentField` maps the body into each cell's frame, sums, and clamps. The spawner mirrors the pooled, seeded gate spawner, so replay, golden, and fairness all hold.

Surge

A seeded scheduler produces an envelope in $[0,1]$ (idle $\rightarrow$ ramp-in $\rightarrow$ hold $\rightarrow$ ramp-out); the surge field is whole-screen turbulence + drift scaled by it, summed *before* the clamp so cells + surge share one escape budget.

Determinism

All fields are pure in (x, y, t) ; spawner and surge run on their own seeded streams — same seed, same run.

§5 Quality & fairness

- **156 unit/sim tests green** (44 new). Typecheck clean. Classic golden re-verified each phase.
- **Fairness by construction:** a single budget clamp anchored to the weakest creature (Otter) — intensity can exceed 1.0 and the emitted force stays escapable.
- **G-Currents spot check (interim):** 60 seeds $\times$ 4 creatures, currents-on Power Dive, reference bot — $\leq 1/60$ failures (~98% clearance). The full 80k sweep + a frozen currents golden is the outstanding formal gate.
- **Bit-identical guard:** currents-off is byte-for-byte identical whether the flag is omitted or explicitly false.

§6 Design forks as sandbox variables

Per direction, the open decisions are variables, not commitments — all in `src/config/currents.json`, tunable live in the Sandbox.

FORK	LIVES IN	DEFAULT
Which modes carry currents	<code>modes.*.cells / .surge</code>	all off
Catalog size / kinds	<code>cells.kinds</code>	12 cell kinds
Cell density	<code>cells.spacing</code>	300 px
Intensity	<code>cells.intensity</code>	1.0
Surge rarity / lethality	<code>surge.*</code>	60–90 s · non-lethal
Escape-budget headroom	<code>budgets.*</code>	0.7 / 0.5 / 0.8

§7 Risk register

#	RISK	SEV	MITIGATION / STATUS
R1	Spot check ≠ full sweep; a rare seed could be unfair	MED	Clamp bounds per-frame force; full 80k sweep + golden tracked
R2	Joint gate + current difficulty untuned (escapable ≠ fun)	MED	Sandbox exists for feel-tuning; defaults conservative
R3	No player-facing entry yet	LOW · INT	Deep-links + dev scenes; menu/mode entry scoped next
R4	Telegraph draw + sampling perf at high density	LOW	Cheap x-cull; simple Graphics; monitor if density raised
R5	Preview panel unreachable this session (shared port)	LOW	Verified via independent Playwright capture

§8 Recommended next steps

- **Merge PR #2** (zero-risk) → Catalog + Sandbox live for review.
- **Feel-tuning pass** in the Sandbox → lock default intensity/spacing.
- **Player entry:** ship a "Currents" toggle or mode so users can actually play it.
- **Formal G-Currents:** full 80k sweep + frozen currents golden.
- **Ratify ADR-014**, then graduate candidate patterns (whirl-pair, shear-step, breathing-wall, roaming-eddy) as each passes the sweep.

Companion: docs/currents/CURRENTS_PROGRAM.md (plan of record) · PR #2 · Floatsam /
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